

APPENDIX A

(REQUIRED FOR OPEN-BOOK EXAMINATION)

DESIGN DATA

BELT DRIVE

TABLES:

- A-1: Cross Section Dimensions and Belt Lengths
- A-2: Service Factor
- A-3: Cross Section Selection Chart
- A-4: Standard Sheave Diameters
- A-5: Belt Length Correction Factor C_L
- A-6: Angle of Contact Correction Factor C_θ
- A-7: Power Ratings - SPZ, SPA, SPB and SPC
- A-8: Sheave Groove Dimensions and Effective Coefficient of Friction

Source: Uni-Drive Transmission Equipment – Roulunds Roflex V-belts

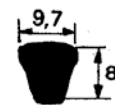
Table A-1: Cross Section Dimensions and Belt Lengths

Section	Dimension W x H	Recommended minimum diameter		Belt weight kg/m
SPZ	9.7 x 8	Pitch diameter d _p	63	0.070
SPA	12.7 x 10		90	0.119
SPB	16.3 x 13		140	0.194
SPC	22 x 18		224	0.360
19	18.6 x 15		180	0.250

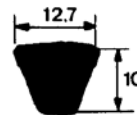
Standard Pitch Length

SPZ	SPA	SPB	SPC
SPZ 630			
SPZ 670			
SPZ 710			
SPZ 750			
SPZ 800	SPA 800		
SPZ 850	SPA 850		
SPZ 900	SPA 900		
SPZ 950	SPA 950		
SPZ 1000	SPA 1000		
SPZ 1060	SPA 1060		
SPZ 1120	SPA 1120		
SPZ 1180	SPA 1180		
SPZ 1250	SPA 1250	SPB 1250	
SPZ 1320	SPA 1320	SPB 1320	
SPZ 1400	SPA 1400	SPB 1400	
SPZ 1500	SPA 1500	SPB 1500	
SPZ 1600	SPA 1600	SPB 1600	
SPZ 1700	SPA 1700	SPB 1700	
SPZ 1800	SPA 1800	SPB 1800	
SPZ 1900	SPA 1900	SPB 1900	
SPZ 2000	SPA 2000	SPB 2000	SPC 2000
SPZ 2120	SPA 2120	SPB 2120	SPC 2120
SPZ 2240	SPA 2240	SPB 2240	SPC 2240
SPZ 2360	SPA 2360	SPB 2360	SPC 2360
SPZ 2500	SPA 2500	SPB 2500	SPC 2500
SPZ 2650	SPA 2650	SPB 2650	SPC 2650
SPZ 2800	SPA 2800	SPB 2800	SPC 2800
SPZ 3000	SPA 3000	SPB 3000	SPC 3000
SPZ 3150	SPA 3150	SPB 3150	SPC 3150
SPZ 3350	SPA 3350	SPB 3350	SPC 3350
SPZ 3550	SPA 3550	SPB 3550	SPC 3550
	SPA 3750	SPB 3750	SPC 3750
	SPA 4000	SPB 4000	SPC 4000
	SPA 4250	SPB 4250	SPC 4250
	SPA 4500	SPB 4500	SPC 4500
		SPB 4750	SPC 4750
		SPB 5000	SPC 5000
		SPB 5300	SPC 5300
		SPB 5600	SPC 5600
		SPB 6000	SPC 6000
		SPB 6300	SPC 6300
		SPB 6700	SPC 6700
		SPB 7100	SPC 7100
		SPB 7500	SPC 7500
		SPB 8000	SPC 8000
		SPB 8500	SPC 8500
		SPB 9000	SPC 9000
			SPC 9500
			SPC 10000
			SPC 10600
			SPC 11200
			SPC 11800
			SPC 12500

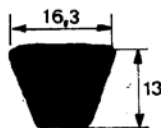
(mm)



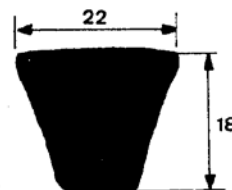
$$\text{SPZ } L_a = L_p + 13 \text{ mm}$$



$$\text{SPA } L_a = L_p + 18 \text{ mm}$$



$$\text{SPB } L_a = L_p + 22 \text{ mm}$$



$$\text{SPC } L_a = L_p + 30 \text{ mm}$$

La = Outside length
Lp = Pitch length

Although the cross sections are identical, some manufacturers use other designations viz: —
 SPZ is interchangeable with Alpha, 3V and 9N
 SPB is interchangeable with Beta, 5V and 15N

Table A-2: Service factors

for different kinds of driver and driven machine combinations are shown below. The machines listed are representative examples only. When the particular machine considered is not listed, select a listed machine which has similar load characteristics.

DRIVEN UNIT	DRIVING UNIT					
	AC Motors, Normal Torque, Squirrel Cage, Synchronous and Split Phase. DC Motors, Shunt-Wound, Multiple Cylinder Internal Combustion Engines.			AC Motors, High Torque, High Slip, Repulsion - Induction, Single Phase, Series Wound and Slip Ring. DC Motors, Series Wound and Compound Wound. Single Cylinder Internal Combustion Engines. Line Shafts. Clutches.		
	Number of daily service hours <10 10-16 16-24			Number of daily service hours <10 10-16 16-24		
Agitators for Liquids, Blowers and Exhausters, Fans up to 7.5 kW, Light Duty Conveyors	1,0	1,1	1,2	1,1	1,2	1,3
Belt conveyors for sand, grain, etc., Bough Mixers, Fans over 7.5 kW, Generators, Laundry Machinery, Machinery, Positive Displacement Rotary Pumps, Revolving and Vibrating Screens Line Shafts	1,1	1,2	1,3	1,2	1,3	1,4
Brick Machinery, Bucket Elevators, Piston Compressors, Conveyors (Drag-Pan-Screw), Hammer Mills, Paper Mill beaters, Piston Pumps, Positive Displacement Blowers, Pulverizers, Saw Mill and Wood-working Machinery, Textile Machinery	1,2	1,3	1,4	1,4	1,5	1,6
Crushers (Gyratory-Jaw-Roll), Mills (Ball-Rod), Moists, Rubber Calenders, Extruders - Mills.	1,3	1,4	1,5	1,5	1,6	1,8

Table A-3: Choice of Cross Section

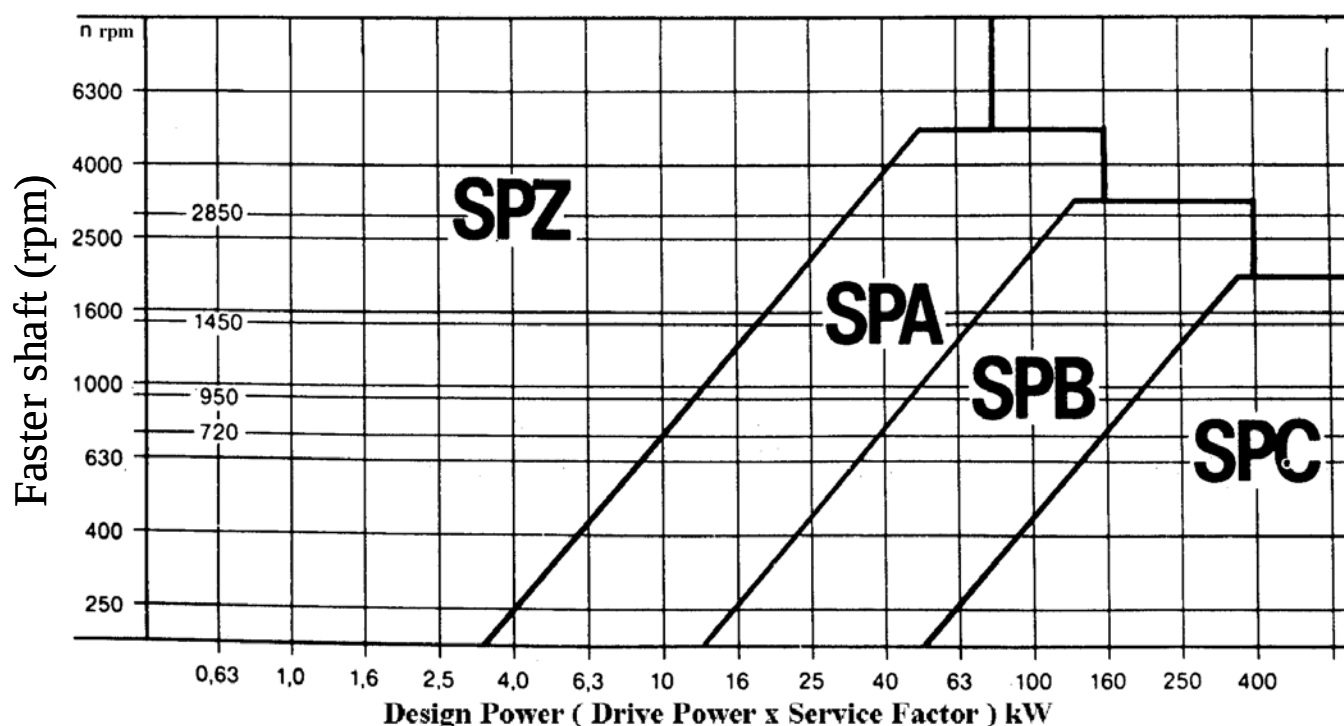


Table A-4: Standard Sheave Diameters (mm)

SPZ/3V	63	71	75	80	90	100	112	125	132	140	150	160	180	200	224	250	280	315	355	400	500	630			
SPA/4V	90	95	100	106	112	118	125	132	140	150	160	180	200	224	250	280	315	355	400	450	500	560	630	710	800
SPB/5V	140	150	160	170	180	200	224	250	280	315	355	400	450	500	560	600	710	750	800	900	1000	1120			
SPC/7V	224	236	250	265	280	300	315	335	355	400	450	500	560	600	630	710	750	800	900	1000	1120	1250	1400	1600	2000

Table A-5: Correction Factor for Arc of Contact C_θ

Arc of contact of smaller sheave	180°	174°	167°	163°	157°	151°	145°	139°	133°	127°	120°	113°	106°	99°	91°
C_θ	1.00	0.99	0.97	0.96	0.94	0.93	0.91	0.89	0.87	0.85	0.82	0.79	0.77	0.73	0.70

Table A-6: Standard Belt Length L (mm) and Belt Length Correction Factor C_L

Profile SPZ		Profile SPA		Profile SPB		Profile SPC	
L (mm)	Correction factor	L (mm)	Correction factor	L (mm)	Correction factor	L (mm)	Correction factor
630	0.85	800	0.81	1250	0.83	2000	0.83
710	0.87	900	0.83	1400	0.85	2240	0.85
800	0.89	1000	0.85	1600	0.87	2500	0.87
900	0.90	1120	0.87	1800	0.89	2800	0.89
1000	0.92	1250	0.89	2000	0.90	3150	0.90
1120	0.94	1400	0.90	2240	0.92	3550	0.92
1250	0.96	1600	0.93	2500	0.94	4000	0.94
1400	0.98	1800	0.94	2800	0.96	4500	0.96
1600	1.00	2000	0.96	3150	0.98	5000	0.98
1800	1.02	2240	0.98	3550	1.00	5600	1.00
2000	1.04	2500	1.00	4000	1.02	6300	1.02
2240	1.06	2800	1.02	4500	1.04	7100	1.04
2500	1.08	3150	1.04	5000	1.06	8000	1.06
2800	1.10	3550	1.06	5600	1.08	9000	1.08
3150	1.12	4000	1.08	6300	1.10	10000	1.10
3550	1.14	4500	1.10	7100	1.12	11200	1.12
				8000	1.14	12500	1.14

Standard belt lengths acc. to ISO 4148

Non-standard belt lengths available on request

Correction factor can be interpolated for intermediate belt lengths.

Table A-7a

kW-ratings per V-belt **P** [kW] at 180° arc of contact**SPZ**

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave													
		200	400	800	1200	1600	2000	2400	3200	3600	4000	4500	5000	5500	6000
63	1.00	0.19	0.33	0.58	0.80	0.99	1.16	1.32	1.58	1.69	1.79	1.89	1.96	2.01	2.03
	1.05	0.19	0.35	0.61	0.84	1.04	1.23	1.39	1.69	1.81	1.92	2.03	2.12	2.19	2.23
	1.20	0.21	0.37	0.66	0.90	1.14	1.35	1.55	1.89	2.04	2.17	2.32	2.44	2.54	2.61
	1.50	0.22	0.39	0.70	0.97	1.22	1.45	1.67	2.05	2.22	2.37	2.54	2.69	2.81	2.91
	≥3.00	0.22	0.40	0.72	1.01	1.27	1.51	1.74	2.14	2.32	2.49	2.67	2.83	2.97	3.08
71	1.00	0.24	0.44	0.79	1.10	1.38	1.63	1.87	2.29	2.47	2.63	2.81	2.95	3.06	3.13
	1.05	0.25	0.46	0.81	1.13	1.43	1.70	1.95	2.39	2.59	2.76	2.95	3.11	3.24	3.33
	1.20	0.26	0.48	0.86	1.21	1.53	1.83	2.10	2.60	2.82	3.02	3.24	3.43	3.59	3.71
	1.50	0.27	0.50	0.90	1.27	1.61	1.93	2.22	2.76	3.00	3.22	3.47	3.68	3.86	4.01
	≥3.00	0.28	0.51	0.93	1.30	1.65	1.98	2.29	2.85	3.10	3.33	3.59	3.82	4.02	4.18
80	1.00	0.31	0.56	1.02	1.43	1.81	2.16	2.48	3.07	3.32	3.55	3.80	4.01	4.17	4.29
	1.05	0.31	0.58	1.04	1.47	1.86	2.22	2.56	3.17	3.44	3.68	3.95	4.17	4.35	4.48
	1.20	0.33	0.60	1.09	1.54	1.96	2.35	2.72	3.38	3.67	3.94	4.24	4.49	4.70	4.86
	1.50	0.34	0.62	1.13	1.60	2.04	2.45	2.84	3.54	3.85	4.14	4.46	4.74	4.98	5.16
	≥3.00	0.34	0.63	1.16	1.64	2.09	2.51	2.90	3.63	3.95	4.25	4.59	4.88	5.13	5.33
90	1.00	0.38	0.70	1.27	1.79	2.28	2.73	3.15	3.91	4.24	4.54	4.86	5.12	5.32	5.46
	1.05	0.39	0.71	1.30	1.83	2.33	2.80	3.23	4.01	4.36	4.67	5.00	5.29	5.50	5.66
	1.20	0.40	0.73	1.35	1.91	2.43	2.92	3.39	4.22	4.59	4.92	5.29	5.60	5.85	6.04
	1.50	0.41	0.76	1.39	1.97	2.51	3.02	3.51	4.38	4.77	5.12	5.52	5.85	6.13	6.34
	≥3.00	0.41	0.77	1.41	2.00	2.56	3.08	3.57	4.47	4.87	5.23	5.64	5.99	6.28	6.51
100	1.00	0.45	0.83	1.52	2.15	2.74	3.30	3.81	4.72	5.12	5.48	5.85	6.16	6.38	6.51
	1.05	0.45	0.84	1.55	2.19	2.80	3.36	3.89	4.83	5.24	5.61	6.00	6.32	6.56	6.70
	1.20	0.47	0.87	1.60	2.27	2.90	3.49	4.04	5.03	5.47	5.86	6.29	6.64	6.91	7.09
	1.50	0.48	0.89	1.64	2.33	2.98	3.59	4.16	5.19	5.65	6.06	6.51	6.89	7.18	7.39
	≥3.00	0.48	0.90	1.66	2.36	3.02	3.65	4.23	5.28	5.75	6.17	6.64	7.03	7.34	7.56
112	1.00	0.53	0.99	1.82	2.58	3.30	3.96	4.58	5.67	6.14	6.54	6.97	7.29	7.50	7.59
	1.05	0.54	1.00	1.84	2.62	3.35	4.03	4.66	5.77	6.25	6.68	7.12	7.45	7.68	7.78
	1.20	0.55	1.02	1.89	2.70	3.45	4.15	4.81	5.98	6.48	6.93	7.40	7.77	8.03	8.16
	1.50	0.56	1.04	1.93	2.76	3.53	4.25	4.93	6.14	6.66	7.13	7.63	8.02	8.30	8.47
	≥3.00	0.57	1.06	1.96	2.79	3.57	4.31	5.00	6.23	6.76	7.24	7.75	8.16	8.46	8.63
125	1.00	0.62	1.16	2.14	3.04	3.88	4.67	5.39	6.65	7.17	7.62	8.07	8.37	8.52	8.51
	1.05	0.63	1.17	2.16	3.08	3.93	4.73	5.47	6.75	7.29	7.75	8.21	8.54	8.70	8.71
	1.20	0.64	1.19	2.21	3.16	4.04	4.86	5.62	6.96	7.52	8.01	8.50	8.85	9.05	9.09
	1.50	0.65	1.21	2.25	3.22	4.12	4.96	5.74	7.12	7.70	8.21	8.72	9.10	9.33	9.39
	≥3.00	0.65	1.23	2.28	3.25	4.16	5.02	5.81	7.21	7.80	8.32	8.85	9.24	9.47	9.55
140	1.00	0.72	1.35	2.50	3.56	4.55	5.46	6.30	7.72	8.29	8.76	9.18	9.42	9.43	
	1.05	0.73	1.36	2.53	3.60	4.60	5.53	6.37	7.82	8.41	8.89	9.33	9.58	9.62	
	1.20	0.74	1.39	2.58	3.68	4.70	5.65	6.53	8.03	8.63	9.14	9.60	9.89	9.97	
	1.50	0.75	1.41	2.62	3.74	4.78	5.75	6.65	8.19	8.82	9.34	9.84	10.15	10.25	
	≥3.00	0.76	1.42	2.64	3.77	4.83	5.81	6.72	8.28	8.92	9.45	9.97	10.29	10.40	
160	1.00	0.86	1.60	2.98	4.24	5.41	6.48	7.45	9.03	9.62	10.06	10.39	10.43		
	1.05	0.86	1.62	3.01	4.28	5.47	6.55	7.53	9.14	9.74	10.20	10.53	10.59		
	1.20	0.88	1.64	3.06	4.36	5.57	6.68	7.68	9.34	9.97	10.45	10.82	10.91		
	1.50	0.89	1.66	3.10	4.42	5.65	6.78	7.80	9.50	10.15	10.65	11.05	11.16		
	≥3.00	0.89	1.67	3.12	4.45	5.69	6.83	7.87	9.59	10.25	10.76	11.17	11.30		
180	1.00	0.99	1.86	3.45	4.91	6.25	7.47	8.54	10.21	10.77	11.12	11.24			
	1.05	1.00	1.87	3.48	4.95	6.31	7.53	8.62	10.31	10.89	11.25	11.39			
	1.20	1.01	1.89	3.53	5.03	6.41	7.66	8.77	10.52	11.11	11.51	11.67			
	1.50	1.02	1.91	3.57	5.09	6.49	7.76	8.89	10.68	11.30	11.71	11.90			
	≥3.00	1.02	1.93	3.59	5.12	6.53	7.82	8.96	10.77	11.40	11.82	12.03			
200	1.00	1.12	2.11	3.92	5.57	7.07	8.41	9.55	11.24	11.71	11.90				
	1.05	1.13	2.12	3.94	5.61	7.12	8.47	9.64	11.34	11.82	12.03				
	1.20	1.14	2.15	3.99	5.69	7.22	8.60	9.79	11.55	12.05	12.28				
	1.50	1.15	2.17	4.03	5.75	7.30	8.70	9.91	11.71	12.23	12.48				
	≥3.00	1.16	2.18	4.06	5.78	7.35	8.76	9.97	11.80	12.34	12.59				

Dynamically balanced
sheaves are requested.

Table A-7b

kW-ratings per V-belt **P** [kW] at 180° arc of contact**SPA**

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave													
		200	400	800	1200	1600	2000	2400	3200	3600	4000	4500	5000	5500	6000
90	1.00	0.42	0.74	1.30	1.77	2.18	2.54	2.85	3.32	3.48	3.57	3.61	3.55	3.38	3.10
	1.05	0.43	0.77	1.36	1.86	2.30	2.70	3.04	3.57	3.75	3.88	3.96	3.94	3.81	3.57
	1.20	0.46	0.83	1.48	2.04	2.54	3.00	3.40	4.05	4.29	4.48	4.63	4.69	4.63	4.46
	1.50	0.49	0.88	1.57	2.18	2.73	3.23	3.68	4.43	4.72	4.96	5.17	5.28	5.28	5.17
	≥3.00	0.50	0.90	1.62	2.26	2.84	3.36	3.84	4.64	4.96	5.22	5.47	5.61	5.65	5.57
100	1.00	0.53	0.96	1.71	2.37	2.96	3.49	3.95	4.69	4.96	5.15	5.28	5.28	5.13	4.84
	1.05	0.55	0.99	1.77	2.47	3.09	3.64	4.14	4.94	5.24	5.46	5.63	5.67	5.56	5.30
	1.20	0.58	1.05	1.89	2.64	3.32	3.94	4.50	5.42	5.77	6.06	6.30	6.41	6.38	6.20
	1.50	0.60	1.10	1.99	2.79	3.51	4.18	4.78	5.80	6.20	6.53	6.83	7.00	7.03	6.91
	≥3.00	0.62	1.13	2.04	2.87	3.62	4.31	4.94	6.01	6.44	6.80	7.13	7.34	7.40	7.31
112	1.00	0.67	1.23	2.21	3.09	3.89	4.60	5.24	6.27	6.66	6.94	7.14	7.16	6.99	6.61
	1.05	0.69	1.26	2.27	3.18	4.01	4.76	5.43	6.52	6.93	7.25	7.49	7.55	7.41	7.07
	1.20	0.72	1.32	2.39	3.36	4.25	5.06	5.79	7.00	7.47	7.84	8.16	8.30	8.24	7.97
	1.50	0.74	1.36	2.49	3.50	4.44	5.29	6.07	7.38	7.90	8.32	8.68	8.89	8.89	8.68
	≥3.00	0.75	1.39	2.54	3.58	4.54	5.43	6.23	7.59	8.14	8.58	8.99	9.22	9.25	9.07
125	1.00	0.82	1.51	2.74	3.86	4.87	5.79	6.60	7.91	8.39	8.72	8.96	8.95	8.67	8.10
	1.05	0.83	1.54	2.81	3.95	4.99	5.94	6.79	8.16	8.67	9.04	9.30	9.33	9.09	8.55
	1.20	0.86	1.60	2.92	4.13	5.23	6.24	7.15	8.64	9.21	9.64	9.97	10.08	9.91	9.46
	1.50	0.89	1.65	3.02	4.27	5.42	6.48	7.43	9.02	9.63	10.12	10.52	10.67	10.56	10.17
	≥3.00	0.90	1.67	3.07	4.35	5.53	6.61	7.59	9.22	9.87	10.38	10.81	11.00	10.93	10.57
140	1.00	0.99	1.83	3.35	4.73	5.99	7.12	8.12	9.70	10.25	10.62	10.80	10.65	10.13	
	1.05	1.00	1.86	3.41	4.82	6.11	7.27	8.31	9.95	10.53	10.93	11.15	11.04	10.56	
	1.20	1.03	1.92	3.53	5.00	6.35	7.57	8.67	10.43	11.07	11.52	11.82	11.78	11.38	
	1.50	1.06	1.97	3.63	5.15	6.54	7.81	8.95	10.81	11.49	12.00	12.35	12.38	12.03	
	≥3.00	1.07	1.99	3.68	5.22	6.64	7.94	9.10	11.02	11.73	12.26	12.65	12.71	12.40	
160	1.00	1.21	2.26	4.15	5.88	7.44	8.84	10.06	11.89	12.46	12.78	12.76	12.26		
	1.05	1.23	2.29	4.22	5.97	7.56	8.99	10.24	12.14	12.74	13.08	13.11	12.65		
	1.20	1.26	2.35	4.34	6.15	7.80	9.29	10.60	12.62	13.28	13.68	13.78	13.39		
	1.50	1.28	2.39	4.43	6.29	7.99	9.53	10.89	13.00	13.71	14.16	14.32	13.99		
	≥3.00	1.29	2.42	4.48	6.37	8.10	9.66	11.05	13.21	13.95	14.42	14.61	14.32		
180	1.00	1.43	2.68	4.95	7.00	8.85	10.49	11.88	13.85	14.35	14.49	14.10			
	1.05	1.45	2.71	5.01	7.09	8.97	10.64	12.07	14.09	14.63	14.80	14.45			
	1.20	1.48	2.77	5.13	7.27	9.21	10.94	12.43	14.57	15.17	15.40	15.13			
	1.50	1.50	2.82	5.22	7.41	9.39	11.18	12.71	14.95	15.59	15.87	15.66			
	≥3.00	1.52	2.84	5.27	7.49	9.51	11.31	12.87	15.16	15.83	16.14	15.96			
200	1.00	1.65	3.10	5.73	8.10	10.22	12.06	13.59	15.54	15.87	15.72				
	1.05	1.67	3.13	5.79	8.19	10.34	12.22	13.78	15.79	16.15	16.03				
	1.20	1.70	3.19	5.91	8.37	10.58	12.52	14.13	16.27	16.69	16.62				
	1.50	1.72	3.23	6.00	8.51	10.77	12.75	14.42	16.64	17.12	17.10				
	≥3.00	1.74	3.26	6.06	8.59	10.88	12.88	14.58	16.86	17.35	17.36				
224	1.00	1.92	3.59	6.65	9.39	11.80	13.85	15.47	17.20	17.17					
	1.05	1.93	3.62	6.71	9.47	11.93	14.00	15.65	17.44	17.45					
	1.20	1.96	3.68	6.83	9.66	12.17	14.30	16.01	17.92	17.99					
	1.50	1.99	3.73	6.93	9.80	12.36	14.54	16.30	18.30	18.42					
	≥3.00	2.00	3.76	6.98	9.88	12.46	14.67	16.46	18.51	18.66					
250	1.00	2.20	4.13	7.64	10.75	13.44	15.65	17.28	18.48						
	1.05	2.21	4.16	7.70	10.84	13.57	15.80	17.46	18.73						
	1.20	2.24	4.22	7.82	11.02	13.81	16.10	17.82	19.21						
	1.50	2.27	4.26	7.91	11.17	14.00	16.34	18.11	19.59						
	≥3.00	2.28	4.29	7.96	11.25	14.10	16.47	18.27	19.80						
280	1.00	2.52	4.73	8.75	12.27	15.23	17.53	19.05	19.24						
	1.05	2.54	4.77	8.80	12.36	15.35	17.68	19.23	19.49						
	1.20	2.57	4.83	8.93	12.54	15.59	17.98	19.59	19.96						
	1.50	2.59	4.87	9.03	12.68	15.78	18.22	19.87	20.34						
	≥3.00	2.60	4.90	9.08	12.76	15.89	18.35	20.03	20.56						
315	1.00	2.89	5.44	10.02	13.97	17.16	19.44	20.63							
	1.05	2.91	5.47	10.09	14.06	17.28	19.59	20.82							
	1.20	2.94	5.53	10.21	14.24	17.52	19.89	21.18							
	1.50	2.96	5.57	10.30	14.38	17.71	20.13	21.46							
	≥3.00	2.98	5.60	10.35	14.46	17.82	20.26	21.62							
Dynamically balanced sheaves are requested.															

Table A-7c

kW-ratings per V-belt **P** [kW] at 180° arc of contact**SPB**

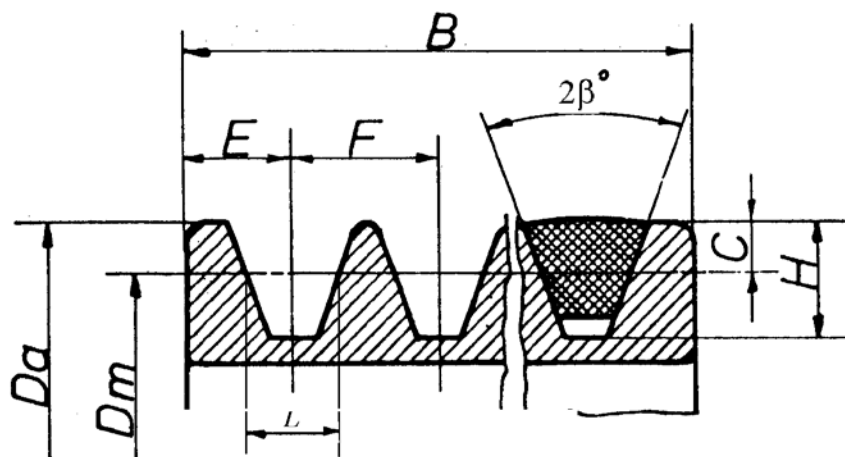
Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave													
		200	400	800	1200	1600	1800	2000	2200	2400	3200	3600	4000	4500	5000
140	1.00	1.20	2.17	3.85	5.30	6.55	7.10	7.60	8.05	8.45	9.43	9.54	9.35	8.68	7.45
	1.05	1.24	2.24	3.99	5.50	6.82	7.41	7.94	8.43	8.85	9.97	10.15	10.04	9.43	8.30
	1.20	1.30	2.37	4.25	5.90	7.35	8.00	8.60	9.14	9.64	11.03	11.34	11.35	10.92	9.93
	1.50	1.35	2.47	4.46	6.21	7.76	8.47	9.12	9.72	10.27	11.87	12.27	12.39	12.09	11.24
	≥3.00	1.38	2.53	4.57	6.38	8.00	8.72	9.41	10.04	10.62	12.33	12.80	12.97	12.74	11.97
160	1.00	1.57	2.86	5.15	7.16	8.91	9.68	10.40	11.03	11.59	13.01	13.15	12.87	11.87	10.08
	1.05	1.60	2.93	5.29	7.36	9.18	9.99	10.74	11.41	12.00	13.55	13.76	13.55	12.64	10.93
	1.20	1.67	3.06	5.55	7.76	9.71	10.59	11.39	12.13	12.79	14.60	14.95	14.87	14.12	12.57
	1.50	1.72	3.17	5.76	8.07	10.12	11.05	11.91	12.70	13.41	15.43	15.88	15.91	15.29	13.87
	≥3.00	1.75	3.22	5.88	8.25	10.36	11.32	12.21	13.02	13.76	15.90	16.41	16.49	15.94	14.60
180	1.00	1.93	3.55	6.44	8.99	11.21	12.19	13.08	13.87	14.56	16.19	16.22	15.67	14.07	
	1.05	1.96	3.61	6.57	9.18	11.48	12.49	13.42	14.25	14.97	16.73	16.83	16.35	14.83	
	1.20	2.03	3.75	6.84	9.58	12.00	13.09	14.08	14.97	15.76	17.78	18.02	17.66	16.31	
	1.50	2.08	3.85	7.05	9.89	12.42	13.55	14.60	15.54	16.38	18.61	18.95	18.70	17.48	
	≥3.00	2.11	3.91	7.16	10.07	12.65	13.82	14.89	15.86	16.73	19.08	19.48	19.29	18.14	
200	1.00	2.28	4.23	7.71	10.78	13.43	14.60	15.64	16.56	17.34	18.95	18.71	17.67		
	1.05	2.32	4.29	7.84	10.98	13.71	14.90	15.98	16.93	17.75	19.49	19.32	18.35		
	1.20	2.38	4.42	8.11	11.37	14.23	15.49	16.64	17.66	18.54	20.54	20.50	19.67		
	1.50	2.43	4.53	8.31	11.69	14.65	15.96	17.16	18.23	19.16	21.37	21.44	20.71		
	≥3.00	2.46	4.59	8.43	11.86	14.88	16.23	17.45	18.55	19.51	21.84	21.96	21.29		
224	1.00	2.71	5.03	9.21	12.87	16.01	17.36	18.55	19.57	20.40	21.65	20.83			
	1.05	2.74	5.10	9.34	13.08	16.28	17.67	18.89	19.94	20.81	22.19	21.44			
	1.20	2.81	5.23	9.60	13.47	16.81	18.26	19.55	20.66	21.60	23.24	22.63			
	1.50	2.86	5.34	9.82	13.78	17.22	18.73	20.07	21.24	22.22	24.07	23.56			
	≥3.00	2.89	5.39	9.93	13.96	17.46	18.99	20.36	21.56	22.57	24.54	24.09			
250	1.00	3.17	5.90	10.81	15.09	18.68	20.18	21.48	22.54	23.35	23.75				
	1.05	3.20	5.97	10.95	15.29	18.95	20.49	21.82	22.91	23.76	24.29				
	1.20	3.27	6.10	11.21	15.69	19.48	21.08	22.47	23.63	24.54	25.34				
	1.50	3.32	6.20	11.42	16.00	19.89	21.55	22.99	24.20	25.17	26.17				
	≥3.00	3.35	6.26	11.53	16.17	20.13	21.81	23.28	24.53	25.52	26.64				
280	1.00	3.69	6.89	12.62	17.56	21.59	23.21	24.54	25.56	26.23	24.99				
	1.05	3.72	6.96	12.76	17.76	21.86	23.52	24.88	25.93	26.64	25.53				
	1.20	3.79	7.09	13.02	18.16	22.38	24.11	25.54	26.65	27.43	26.59				
	1.50	3.84	7.19	13.23	18.47	22.80	24.57	26.06	27.22	28.05	27.42				
	≥3.00	3.87	7.25	13.35	18.64	23.03	24.84	26.35	27.55	28.40	27.88				
315	1.00	4.29	8.03	14.69	20.32	24.73	26.39	27.66	28.48	28.82					
	1.05	4.33	8.10	14.83	20.53	25.00	26.70	28.00	28.85	29.23					
	1.20	4.39	8.23	15.09	20.92	25.53	27.29	28.65	29.57	30.02					
	1.50	4.45	8.33	15.30	21.23	25.94	27.76	29.17	30.15	30.64					
	≥3.00	4.48	8.39	15.42	21.41	26.17	28.02	29.46	30.47	30.99					
355	1.00	4.98	9.32	17.00	23.31	27.96	29.54	30.55	30.94						
	1.05	5.01	9.38	17.13	23.52	28.23	29.84	30.89	31.31						
	1.20	5.08	9.52	17.40	23.91	28.76	30.43	31.54	32.04						
	1.50	5.13	9.62	17.60	24.22	29.17	30.90	32.06	32.61						
	≥3.00	5.16	9.68	17.72	24.40	29.41	31.16	32.36	32.93						
400	1.00	5.74	10.74	19.51	26.45	31.09	32.37	32.86							
	1.05	5.78	10.81	19.64	26.66	31.37	32.68	33.20							
	1.20	5.84	10.94	19.91	27.05	31.89	33.27	33.85							
	1.50	5.90	11.05	20.11	27.36	32.31	33.74	34.37							
	≥3.00	5.92	11.11	20.23	27.54	32.54	34.00	34.67							
		Dynamically balanced sheaves are requested.													

Table A-7d

kW-ratings per V-belt **P** [kW] at 180° arc of contact**SPC**

Pitch diam. of smaller sheave [mm]	$\frac{D_{BIG}}{D_{SMALL}}$ ratio	RPM of smaller sheave													
		200	300	400	500	600	800	1200	1600	1800	2000	2200	2400	2600	3200
224	1.00	3.63	5.15	6.58	7.94	9.22	11.64	15.77	18.94	20.11	20.99	21.56	21.78	21.64	18.78
	1.05	3.72	5.29	6.77	8.17	9.51	12.01	16.33	19.68	20.95	21.92	22.58	22.89	22.84	20.26
	1.20	3.90	5.56	7.13	8.62	10.05	12.73	17.41	21.12	22.57	23.72	24.56	25.05	25.18	23.14
	1.50	4.05	5.77	7.41	8.97	10.48	13.30	18.27	22.26	23.85	25.15	26.13	26.76	27.03	25.42
	≥3.00	4.13	5.89	7.57	9.18	10.72	13.62	18.74	22.90	24.57	25.95	27.00	27.72	28.07	26.70
250	1.00	4.48	6.38	8.19	9.89	11.54	14.61	19.89	23.90	25.37	26.44	27.08	27.27	26.95	22.68
	1.05	4.57	6.52	8.37	10.14	11.82	14.98	20.44	24.64	26.21	27.37	28.11	28.38	28.16	24.17
	1.20	4.75	6.79	8.72	10.59	12.36	15.70	21.52	26.08	27.83	29.17	30.09	30.54	30.50	27.05
	1.50	4.89	7.00	9.02	10.94	12.79	16.27	22.38	27.22	29.11	30.59	31.65	32.25	32.35	29.33
	≥3.00	4.97	7.12	9.18	11.14	13.03	16.59	22.86	27.86	29.83	31.39	32.53	33.21	33.39	30.60
280	1.00	5.44	7.79	10.02	12.14	14.18	17.98	24.48	29.31	31.00	32.15	32.71	32.63	31.87	
	1.05	5.54	7.93	10.20	12.38	14.46	18.35	25.04	30.05	31.84	33.08	33.73	33.75	33.08	
	1.20	5.72	8.20	10.56	12.83	15.00	19.07	26.12	31.49	33.46	34.88	35.71	35.91	35.42	
	1.50	5.86	8.41	10.85	13.18	15.42	19.64	26.97	32.63	34.74	36.30	37.28	37.61	37.27	
	≥3.00	5.94	8.53	11.01	13.38	15.66	19.96	27.45	33.27	35.46	37.10	38.15	38.57	38.31	
315	1.00	6.56	9.42	12.13	14.72	17.20	21.83	29.62	35.16	36.93	37.95	38.15	37.45	35.80	
	1.05	6.66	9.55	12.32	14.96	17.48	22.20	30.18	35.90	37.76	38.88	39.17	38.57	37.01	
	1.20	6.84	9.83	12.68	15.41	18.02	22.92	31.26	37.34	39.38	40.68	41.15	40.73	39.35	
	1.50	6.98	10.04	12.96	15.76	18.45	23.49	32.11	38.48	40.67	42.10	42.71	42.44	41.20	
	≥3.00	7.06	10.16	13.12	15.96	18.69	23.81	32.59	39.12	41.38	42.90	43.59	43.40	42.24	
355	1.00	7.83	11.26	14.52	17.63	20.60	26.11	35.19	41.17	42.78	43.33	42.73			
	1.05	7.92	11.40	14.71	17.86	20.88	26.48	35.75	41.92	43.62	44.26	43.75			
	1.20	8.10	11.67	15.07	18.31	21.42	27.20	36.83	43.36	45.23	46.06	45.73			
	1.50	8.25	11.88	15.35	18.67	21.85	27.77	37.68	44.50	46.52	47.48	47.29			
	≥3.00	8.33	12.00	15.51	18.87	22.09	28.09	38.16	45.13	47.23	48.28	48.17			
400	1.00	9.24	13.31	17.17	20.85	24.35	30.77	41.03	47.01	48.06	47.63				
	1.05	9.34	13.45	17.36	21.08	24.62	31.15	41.59	47.75	48.90	48.56				
	1.20	9.52	13.72	17.72	21.53	25.16	31.87	42.67	49.19	50.52	50.36				
	1.50	9.66	13.93	18.00	21.89	25.59	32.44	43.52	50.33	51.80	51.78				
	≥3.00	9.74	14.05	18.16	22.09	25.83	32.75	44.00	50.97	52.52	52.58				
450	1.00	10.80	15.56	20.07	24.35	28.41	35.76	46.95	52.22	52.14					
	1.05	10.89	15.70	20.26	24.59	28.69	36.13	47.51	52.96	52.97					
	1.20	11.07	15.97	20.62	25.04	29.23	36.85	48.59	54.40	54.59					
	1.50	11.22	16.18	20.90	25.39	29.65	37.42	49.44	55.54	55.87					
	≥3.00	11.30	16.30	21.06	25.59	29.89	37.74	49.92	56.18	56.59					
500	1.00	12.34	17.78	22.93	27.79	32.36	40.53	52.21	55.94						
	1.05	12.44	17.92	23.11	28.02	32.64	40.90	52.77	56.68						
	1.20	12.62	18.19	23.47	28.47	33.18	41.62	53.85	58.12						
	1.50	12.76	18.40	23.76	28.83	33.60	42.19	54.70	59.26						
	≥3.00	12.84	18.52	23.92	29.03	33.84	42.51	55.18	59.90						
560	1.00	14.17	20.41	26.29	31.82	36.95	45.94	57.58	58.25						
	1.05	14.27	20.55	26.48	32.05	37.23	46.31	58.13	58.99						
	1.20	14.45	20.82	26.84	32.50	37.77	47.03	59.21	60.43						
	1.50	14.59	21.04	27.13	32.85	38.20	47.60	60.07	61.57						
	≥3.00	14.67	21.16	27.29	33.05	38.44	47.92	60.55	62.21						
630	1.00	16.29	23.44	30.14	36.37	42.09	51.79	62.40							
	1.05	16.38	23.58	30.33	36.60	42.37	52.16	62.96							
	1.20	16.56	23.85	30.69	37.05	42.91	52.88	64.04							
	1.50	16.70	24.06	30.97	37.41	43.34	53.45	64.89							
	≥3.00	16.78	24.18	31.13	37.61	43.58	53.77	65.37							
710	1.00	18.68	26.84	34.42	41.39	47.66	57.80	65.81							
	1.05	18.77	26.98	34.61	41.62	47.94	58.17	66.37							
	1.20	18.95	27.25	34.97	42.07	48.48	58.89	67.45							
	1.50	19.09	27.46	35.25	42.42	48.91	59.46	68.30							
	≥3.00	19.17	27.58	35.41	42.62	49.15	59.78	68.78							
		Dynamically balanced sheaves are requested.													

Table A-8

TAPER LOCK PULLEY

The pulleys conform to
ISO 4183 – 1980 and DIN 2211

Designed for use with both
Wedge (narrow) and
classical V-belts

$$B = F (n - 1) + 2 E$$

n = No. of Grooves

Dimensions in millimetres

Profile	Dm	2β	L	E	F	C	H	Effective Coeff Friction, f_e
SPZ	≤ 80	34	8,5	8	12	2,00	11	0.45
	> 80	38						0.40
SPA	≤ 118	34	11	10	15	2,75	13,75	0.45
	> 118	38						0.40
SPB	≤ 190	34	14	12,5	19	3,50	17,50	0.45
	> 190	38						0.40
SPC	≤ 315	34	19	17	25,5	4,8	24	0.45
	> 315	38						0.40

Material:

Fine grain cast-iron – Grade 260–200

Eccentricity of O/D to bore:

DIN 2211 part 1

Groove side wobble tolerances:

DIN 2211 part 1

Surface roughness:

ISO 254 – 1981

Balance quality grade:

ISO 1940 – 1973

Pulleys with d_m (pcd) > 125 mm are static
balanced to grade q 6,3.

A residual unbalance of 2 g on the pulley
 d_m (pcd) will be accepted.

The residual unbalance is ≤ 100 g mm/kg
at 600 rev./min. and 20 g mm/kg at
3000 rev./min.

Dynamic balancing can be supplied to
order.